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10 June 1970
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MILITARY SPECIFICATION

POWER SUPPLY, TYPE IV(A)

(TORPEDO TESTING)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement of a rectifier type power supply for use in Naval shipboard or shore-based torpedo workshops. Power Supply Type IV(A) will supply various dc voltages for torpedo electrical systems normally energized by storage batteries. In addition to supplying continuous direct current, it must also be capable of providing periodic current pulses of relatively large magnitude and short duration.

1.2 Classification. The power supply shall be of the following one type :

(a) Input voltage: 208/220/440 volts, 50/60 HZ, 3-phase ac.

(b) Output rating:

(1) Regulated, continuously variable from 6 volts dc or less to 40 volts dc or greater with a 100 ampere continuous capacity and a 200 ampere capacity from start to 15 seconds.

(2) Regulated, continuously variable from 6 volts dc or less to 90 volts dc or greater with a 50 ampere continuous capacity and a 100 ampere capacity from start to 15 seconds.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 1355

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SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed, and Lockcorner

MILITARY

MIL-E-1	Electron Tubes and Crystal Rectifiers
MIL-C-5	Capacitors, Fixed, Mica-Dielectric General Specification for
MIL-R-11	Resistors, Fixed, Composition (Insulated)
MIL-R-26	Resistors, Fixed, (Wirewound, Power Type)
ML-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electrical, AN Type
MIL-A- 5092	Adhesive, Rubber Base, General Purpose
MIL-M-103O4	Meters, Electrical-Indicating, Panel Type, Ruggedized, General Specification for
MIL-W-10434	Windows, Protective Envelope (For Installation in Water-Vaporproof Barriers)
MIL-G-12803	Gasket Material, Nonmetallic
MIL-R-19365	Resistors, Adjustable, Wirewound, Power

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MS20003	Indicator, Humidity, Card, Three Spot, Impregnated Areas

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(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. Power Supply Type IV(A) is of the metallic rectifier type and rectifies 208/220 or 440 volts, 50 or 60 HZ, 3 phase input to direct current of variable regulated potentials. The output voltages are to be selected by controls on the front of the unit. Regulation is to be accomplished by means of transistors, magnetic amplifiers, silicon controlled rectifiers, or a combination of these devices.

3.2 General

3.2.1 Contractor-furnished data. Data required by this specification, or by applicable documents referenced in Section 2, need not be furnished unless specified in the contract or order (see 6.2.10).

3.2.2 Reproduction sample. Unless otherwise specified in the contract or order, two power supplies shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing power supplies that comply with the technical requirements of the contract. See 4.3.3.

3.2.3 Materials. The materials, parts, and assemblies used in the construction of the equipment shall be of a quality consistent with the specified use and performance of the equipment. The largest possible proportion of the material used shall be nonflammable. A design which tends to minimize size and weight shall be employed to the greatest practicable extent.

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3.2.4 Size and weight. The overall dimensions and weight of the power supply shall not exceed the following

Height (in.)	Width (in.)	Depth (in.)	(Weight (lbs)
36	24	24	500

3.2.5 Reliability and safety. During manufacture the contractor shall employ methods that will ensure a quality item having maximum reliability. In the functional application of parts to equipment circuits, adequate factors of safety shall be provided. The design shall include all possible features which will result in stable and reliable operation with reduced requirements for internal adjustment and maintenance.

3.2.6 Maintenance. Accessibility and serviceability features which will lead to simplified maintenance shall be of prime consideration in the design of the power supply.

3.2.7 Cables, connectors, and cable assemblies. Cables, connectors, and cable assemblies shall furnish satisfactory electrical connections and shall be so arranged and wired that leads are not terminated in pins or other exposed contacts that might be accidentally shorted or touched.

3.2.7.1 Output termination. The output receptacle shall be a type MS-3102A-24-9S connector with a mating plug provided. The receptacle shall be mounted on the front panel or access door at least six inches from the bottom of the unit and shall be suitably protected against accidental personnel contact or short circuit. The polarity of this output connector shall be negative at Pin A and positive at pin B.

3.2.8 Controls. The various controls shall operate freely and smoothly without binding or lost motion. Switches, levers, and rotary controls shall be of a design sufficiently rugged to withstand repeated usage. Where stops are used they shall be rugged enough to prevent damage to the mechanism under control.

3.2.8.1 Accessibility. Controls necessary for the operation of the power supply shall be readily accessible, suitably grouped and marked, and of a size and constriction that will permit convenience and ease of operation.

3.2.9 Wiring. All wires shall be of a cross section and temper that till provide ample and safe current carrying capacity and mechanical strength.

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3.2.9.1 Whenever wires are run through holes in metal partitions, shields, and the like; the holes shall be equipped with suitable grommets. Care should be exercised in the running of wire to ensure that it is not carried over or bent around any sharp corner or edge which might, in time, cut through the insulation. To prevent deterioration of the conductors by heat, care shall be taken to ensure that wiring is not exposed to excessive temperatures and that they cannot come in direct contact with heated parts. Special caution should be exercised to ensure that any cabling to components mounted on access doors is of sufficient length to prevent contact with the edge of door at maximum swing.

3.2.9.2 Conductors shall be bound into a cable wherever possible and securely held by means of cord or other means acceptable to the procuring activity.

3.2.10 Wire and component identification. Each conductor shall be identified by color coding or other permanent marking that shall appear at least once in every four inches of wire length. Components shall be identified by circuit symbols or designations marked on or adjacent to the individual components in the unit.

3.2.11 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28.

3.2.12 Conflicting requirements. Conflicting requirements arising between this specification or any specifications or publications listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2.13 Controls and indicators. The following devices shall be mounted on the front of each power supply:

(a) A 4-1/2 inch, dc voltmeter with a single range of 0 to 100 volts, which shall be in accordance with MIL-M-10304 (MR44W100DCVVR). Scale numerals shall be at least 3/16″ high.

(b) A 4-1/2 inch, dc ammeter with a single range of 0 to 100 amperes, which shall be in accordance with MIL-M-10304 (MR44W100DCAAR). Scale numerals shall be at least 3/16" high.

(c) Voltage controls capable of adjusting the output voltage from 6 volts or less to 90 volts or more with incremental changes of less than one-half volt.

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(d) ON and OFF pushbuttons for controlling ON - OFF operation of the power supply's 208/220 or 440 vac input. A motor-starter with manual pushbutton control and fused in the three ac input lines to the power supply. The fuseholders should be separate from the pushbutton switch and easily accessible from the front panel. Fifteen-ampere fuses should be used with a 440 vac input, while 30-ampere fuses should be used with a 220/208 vac input.

(e) An indicator light to indicate "power on" in the ac input circuit. Connection of the indicator light shall be made to the rectifier transformer in such a way that changing the input voltage from 440 to 208 or vice-versa shall not require any change in connection of the indicator light for proper indication.

(f) Voltage range selector: A two-position rotary switch for selecting a LOW or HIGH voltage output range.

(g) Output connector.

3.2.14 Audible frequencies. The design of the power supply shall be such as to minimize the generation of audible frequencies (hum).

3.2.15 Finish. The front panel, frame, chassis, and cabinet shall have a gray baked enamel or plated industrial finish acceptable to the procuring activity.

3.2.16 Enclosures. The power supply enclosures shall be designed with a front access door on which all meters and controls shall be mounted, with the possible exception of the output connector. The enclosure shall be so designed to permit stacking two units, one upon the other, with provisions for hoisting the unit by means of removable eyebolts in its top. Design of the enclosure shall be subject to approval of the procuring activity.

3.2.17 Technical manual. A manual shall be supplied for each power supply and shall contain the theory of operation, schematic diagrams, installation instructions, operating information, maintenance and trouble-shooting procedures, and a parts list. The instruction manual shall be packed with each unit. A recommended spare parts list shall also be included. Contents of the manual shall be subject to the approval of the procuring activity.

3.2.18 Installation data. The contractor will provide the procuring activity with a reproducible copy of basic installation data within 90 days after receipt to the contract. This data shall give all information necessary for planning the installation of power supplies, such as

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overall size, weight, position and type of input connector, power rating, type of ventilation, location of ventilation louvers, control location, and position of output connector. This data shall be shown on the contractor's drawings.

3.3 Performance requirements and product characteristics. The power supply shall meet the following performance requirements and product characteristics:

3.3.1 Input power source. The unit shall operate satisfactorily from any one of the following power sources:

3.3.1.1 Input voltage. The power supply shall operate satisfactorily from 3-phase input voltages of either 208 plus or minus 10 percent, 220 plus or minus 10 percent, or 440 plus or minus 10 percent.

3.3.1.2 Input frequency. 50 or 60 Hz plus or minus 5 percent.

3.3.1.3 Input current. The input current to the power supply shall not exceed 25 amperes per phase when operated at the minimum specified input of 208 minus 10 percent volts while providing rated dc output of 90 volts at 50 amperes. When operating with an input of 440 minus 10 percent volts while providing rated dc output of 90 volts at 50 amperes, the input current to the power supply shall net exceed 12 amperes per phase.

3.3.1.4 Phase sequence. The power supply shall operate as specified without regard to the phase sequence of the input power.

3.3.2 Output voltages and current. The dc output of the power supply shall be available in two ranges, identified as the LOW range and HIGH range, which shall be selectable from the front panel by means of a two-position, rotary-type switch. The voltage and current requirements in both ranges are as follows:

SWITCH POSITION	VOLTAGE RANGE	RATED CONT. CURRENT	MAX. SURGE CURRENT (START TO 15 SECOND)
Low	6 - 40 vdc	100 amperes	200 amperes
HIGH	6 - 90 vdc	50 amperes	100 amperes

3.3.2.1 Static load regulation. With constant line voltage, the output voltage shall not vary more than plus or minus one percent with an output current change fron no-load to continuous rated full load.

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3.3.2.2 Dynamic regulation and response time. In the LOW RANGE position with the sudden application or release of any load current up to 50 amperes, the voltage at the output terminals shall not fall nor rise more than 5 percent from the no-load setting and shall decline to 37 percent of the voltage excursion within 75 milliseconds. This requirement applies to any output voltage above 25 volts. At no-load settings below 25 volts, with the sudden application or release of any load current up to 50 amperes, the voltage at the output terminals shall not fall nor rise more than 1.25 volts and shall recover to within 0.5 volt of the no-load voltage within 75 milliseconds.

3.3.2.2.1 Overshoot or undershoot. A momentary interruption of the ac input power shall not cause the output voltage to overshoot the preset value by more than 5 percent. When ac input power is applied to the power supply, the dc output voltage shall rise at a rate of not less than 30 volts per second to the voltage set by the output voltage control. The output voltage shall not overshoot the preset value by more than 5 percent, and shall return to the preset value in less than 0.5 second. A sudden change in the position of the output voltage control setting shall not cause an overshoot or an undershoot of voltage in excess of 5 percent of the new preset value.

3.3.2.3 Output stability. At any given voltage setting within the voltage range specified in 3.3.2, the output voltage shall remain within plus or minus one volt, without adjustment, for a minimum of four hours, monitored continuously by use of the voltmeter recorder of 4.5.5.

3.3.2.4 Static line regulation. The static line regulation shall not exceed 1 percent for a plus or minus 10 percent line voltage change or a plus or minus 5 percent variation of input frequency at constant load.

3.3.2.5 Output voltage control. Changes in the power supply's output voltage control shall result in simultaneous changes in the dc output voltage, with negligible lag. The voltage control shall be capable of adjusting the output voltage in the range of 6 to 90 volts within increments of less than one-half volt. The output voltage shall be linear within plus or minus 20 degrees with respect to the angular displacement of the output voltage control.

3.3.2.6 Ripple voltage. The ripple content of the dc output of the power supply shall not exceed 0.50 percent rms (1.50 percent peak-to-peak) at any output above 25 volts or current within rating in the LOW voltage range.

3.3.2.7 Output isolation. output of the power supply shall be isolated from ground with either polarity capable of being grounded.

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3.3.2.8 Loss-of-phase protection. Protection against damage to the power supply due to the loss of any one of the 3-phase inputs shall be provided.

3.3.2.9 Series operation. The power supply shall be so designed and constructed that the outputs of two units may be connected series-additive to obtain an output voltage equal to the sum of the output voltages of the individual power supplies. For series operation, no interconnection except the series connection of output terminals shall be required.

NOTE

When operating two power supplies with series-connected output circuits, the output voltage controls shall be varied simultaneously to approximately balance the voltage and load between the two units.

3.3.2.10 Current limiting. The power supply shall incorporate a continuous current limiting circuit to limit the current to 115 amperes plus 0 minus 15 amperes in the HIGH voltage range, and 230 amperes plus 0 minus 30 amperes in the LOW voltage range, and shall automatically restore the output voltage to its original preset value upon removal of the overload.

3.3.2.11 The power supply shall operate as specified herein for any combination of resistive. reductive, capacitive, or rotating inverter (direct current to alternating current motor-generator set) load within the specified load rating of the power supply. Specifically, it should be compatible with Torpedoes Mk 37, 44, 45, and 46 and their associated test equipment.

3.3.2.12 Operating life. The power supply shall have a minimum operating life of 1000 hours, when cycled as described in 4.7.4, with only minor maintenance or repair.

3.3.2.13 Operating temperature. With the unit operating in an ambient temperature of 77 degrees Farenheit (i), no exposed parts of the equipment, including the enclosure, shall under any condition of operation achieve a temperate in excess of 130 degrees F. The front panels and operating controls shall not exceed 100 degrees F.

3.4 Environmental requirements

3.4.1 [illegible] Power Supply Type IV(A) shall meet all the performance [illegible] and product characteristics of this specification, when operated [illegible] temperature range of 30 to 130 degrees F (ambient).

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3.4.2 Humidity The power supply shall meet all the performance requirements and product characteristics of this, specification while operating at relative humidities ranging up to 95 percent at temperatures up to 130 degrees F.

3.4.3 Rough handling. The power supply, packaged and packed as specified in 5.1.1 and 5.2.1, shall meet all performance requirements and product characteristics of this specification after being subjected to the rough handling tests of 4.7.3.

3.5 Cleanup. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash, shall be removed.

3.6 Workmanship. The power supply, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, impregnation of coils, plating, painting, machine-screw assemblage, welding, riveting, brazing, and freedom of parts from all burrs and sharp edges.

The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of' the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of Power Supply Type IV(A) shall be classified as follows:

(a) Quality conformance inspection (paragraph 4.6)

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(b) Preproduction inspection (paragraph 4.7)

(c) Periodic production inspection

4.3 Sampling

4.3.1 For nondestructive examination. assembled equipment shall be selected in each unit of the sample shall be inspected in accordance with 4.4. Any equipment containing one or more defects shall be rejected. If the number of such defective equipments in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided all equipments in the lot have been reinspected by the manufacturer for the defect(s) causing rejection and all noncomplying units in the lot have been either corrected or replaced. Defective equipment of accepted lots will be accepted as part of the lot only after correction of all noncompliance items.

TABLE I

SAMPLING FOR ACCEPTANCE

Lot Size	Sample Size	Acceptance Number Defective	Rejection Number Defective
1-8	All	-	-
9-15	8	0	1
16-25	10	0	1
26-40	13	0	1
41-65	17	0	1
66-110	25	1	2
111-180	28	1	2
181 and over	50	1	2

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the basic order or contract. All units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Preproduction. A preproduction sample of two power supplies, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4. at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no

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defects in the sample. Further production of the power supplies by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.4 Periodic production sample. From each succeeding lot or grand lot of 100 or less units, preserved and packaged as specified in Section 5, two power supplies shall be randomly selected. Unless otherwise specified in the contract or purchase order, these units shall be submitted to an activity designated by The procuring activity to determine compliance of the samples with the requirements of the contract, specifications, and drawings. The power supplies shall be considered satisfactory if they meet all specifications and pertinent drawing requirements and withstand the tests specified in 4.6 and 4.7. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity. Periodic production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.4 Product examinations. Each power supply selected in accordance with Table I shall be subjected to a thorough examination to ascertain that the material, workmanship, assembly, and general appearance of the units are in compliance with the requirements of 3.2.3 through 3.2.11. Inspection shall be accomplished prior to assembly when necessary.

4.5 Test equipment

4.5.1 Power source. A 3-phase power source of the following characteristics:

(a) Voltage capable of being baried over the range of 187 to 242 volts and 396 to 484 volts.

(b) Frequency variable from 57 to 63 Hz and 47-1/2 to 52-1/2 Hz.

(c) Minimum current capability of 25 amperes in each phase.

4.5.2 A metered load capable of being varied from 0-230 amps over a voltage range of 6 to 90 volts direct current (vdc).

4.5.3 Oscilloscope capable of calibration to read 1.00 volt with an accuracy of 5 percent and time up to 100 milliseconds with an accuracy of 5 percent.

4.5.4 A 500 volt dc megger, AN/PSM 1 or equivalent.

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4.5.5 Voltmeter recorder capable of recording voltages from 0 to 100 volts dc versus time, with an accuracy of at least 0.5 percent for a period of at least four hours.

4.5.6 A dc voltmeter capable of measuring 0 to 100 volts dc with an error not greater than one-half percent.

4.5.7 An ac voltmeter capable of measuring 0 to 300 and 0 to 600 volts at 50 or 60 Hz with an accuracy of one percent.

4.6 Quality conformance inspection

4.6.1 Each Power Supply Type IV(A) delivered under contract (with the exception noted in 4.6.3.6) shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection. A list of proposed tests, test procedures, and test equipment shall be furnished by the contractor to the Government inspector for approval by the contracting agency prior to setting up test equipment for quality conformance testing.

4.6.2 Test conditions. Unless otherwise specified, the power supply shall be subjected to quality conformance tests under the following conditions:

4.6.2.1 Temperature. Room ambient 65 to 95 degrees F.

4.6.2.2 Altitude. Normal ground.

4.6.3 Test procedure

NOTE

Unless otherwise specified for acceptance testing, the nominal voltage of 208 volts, 60 Hz, 3-phase shall be used.

4.6.3.1 Connect the power supply to the power source of 4.5.1. Connect the output to the variable load (4.5.2).

4.6.3.2 Energize the power supply and set the input voltage to 187 volts at nominal frequency. Apply a load of 50 amperes at 90 volts and measure the input current of each phase. The readings shall not exceed 25 amperes per phase.

4.6.3.3 set the input voltages and frequency nominal setting. With no load on the output, vary the output voltage from 6 to 90 volts dc.

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Repeat this with half load and with full rated load. The unit must meet the requirements of 3.3.2. This test shall be repeated with nominal voltage inputs of 220 and 440.

4.6.3.4 Static load regulation. Set output voltages at 25 volts dc no-load, and vary the load from no-load to continuous rated full load, recurding the highest and lowest output voltage readings over the range. Calculate regulation as follows:

$$\text{Percentage load regulation} = \frac{\triangle E}{E} \times 100$$

Where: $\triangle E$ = Greatest deviation from no-load output voltage.

E = No-load output voltage.

Repeat test with 50 volts dc output voltage at no load. In both cases the percent load regulation shall not exceed one percent.

4.6.3.5 Dynamic regulation and response time. With the voltage switch in the LOW RANGE position, energize the unit, and with the power source (4.5.1) set at nominal voltage and frequency, set the output voltage at 40 volts dc, no load. Connect the oscilloscope (4.5.3) across the output. Simultaneously, observe the dc output on the oscilloscope, and switch a 50 amp load to the output. Measure the dip in the output voltage trace and the time for the output voltage to return to 37 percent of its excursion value. The output voltage dip shall not exceed 2 volts (5 percent of 40 vdc), and the voltage shall return to 37 percent of the excursion value within 75 milliseconds. Repeat this test with the no-load output voltages at 25 volts dc. The excursion shall be not more than 1.25 volts and shall return to within 1/2 volt of the no-load voltage within 75 milliseconds.

4.6.3.5.1 Overshoot or under shoot. With the voltage switch in the HIGH RANGE position and the oscilloscope (4.5.3) connected across the output, set the output voltage to 40 volts dc no load. Interrupt the input power source (4.5.1) for 100, 250, 500, and 1000 ms intervals. Observe the output voltage on the oscilloscope. It shall not overshoot the 40 volts dc by more than 2 volts peak. Repeat the test with half-load (4.5.2) applied to the output. Set the output to 90 volts. Turn the power supply off. Simultaneously turn the power on and observe the output voltage on the oscilloscope. The output voltage shall reach 90 vdc within 3 seconds, shall not overshoot 90 vdc by more than 4.5 volts and shall return to the preset value in less than 0.5 of a second. After stabilization, decrease the output voltage setting from 90 vdc to approximately 30 vdc within a time span of 2 seconds, observing the oscilloscope at the same time. The undershoot shall be less than 1.5 volts.

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4.6.3.6 Output stability. This test should be conducted only on one of every ten production units. Energize the unit with the power source (4.5.1) set at nominal voltage and frequency. Set the output at 50 volts dc with half load. Connect the voltmeter recorder (4.5.5) to the autput of the unit. Record continuously the output voltage vs. the for a period of four hours and without adjustment of the voltage setting. The output voltage shall remain constant within plus or minus one volt.

4.6.3.7 Static line regulation. Energize the unit with the power source (4.5.1 set at 208 volts, 60 Hz. Set the output at 25 volts dc and half load.

4.6.3.7.1 Input voltage. Vary the input from 187 volts to 229 volts, and record the output voltage. Calculate the line regulation as follows:

$$\text{Percent line regulation (input voltage)} = \frac{\triangle E}{E} \times 100$$

Where: $\triangle E$ = The largest deviation from the initial voltage output (E).

E = Initial voltage output with 208 volts input at half load. The percent line regulation shall not exceed one percent.

4.6.3.7.1.1 Repeat 4.6.3.7.1, varying the input voltage from 198 to 242 volts ac, with the initial half-load setting at 220 volts.

4.6.3.7.2 Input frequency. With 208 volts applied to the input, set the frequency to 60 HZ. Set 25 volts dc output With half load. Vary the input frequency from 57 to 63 Hz and record the output voltage. Calculate the line regulation as follows.

$$\text{Percent line regulation (frequency)} = \frac{\triangle E}{E} \times 100$$

Where: $\triangle E$ = The largest deviation from E.

E = Initial output voltage at 60 HZ input frequency. The line regulation shall not exceed one percent.

4.6.3.7.2.1 Repeat 4.6.3.7.2, varying the frequency from 47.5 to 52.5 Hz and with the initial half-load output voltage setting made at 50 Hz.

4.6.3.7.3 Repeat 4.6.3.7.1, 4.6.3.7.1.1, 4.6.3.7.2, and 4.6.3.7.2.1 with the outputs set at half load and 50 vdc. The line regulation shall not exceed one percent.

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4.6.3.5 Ripple voltage. With an input of 208/220 volts at 50 Hz to the power supply, ripple voltage measurements shall be made at no load, one-half load, and full load, as described below. Take readings at output voltages of 25 vdc and 40 vdc in the LOW voltage range.

(a) The effective root mean square (rms) value of the ripple voltage shall be determined by taking peak-to-peak voltage readings with a properly calibrated oscilloscope (4.5.3), using the largest reading as representing the peak-to-peak value of the ripple voltage. The rms value shall then be determined by dividing this measured peak-to-peak value by three

(b) The value of the output voltage shall be determined by a measurement made with the dc voltmeter (4.5.6) connected across the output voltage terminals.

(c) The measurements above shall be taken and recorded at no load, one-halfload, and full load. The percent ripple voltage shall be recorded for each condition from calculations as follows:

$$\text{Percent ripple voltage} = \frac{\text{rms of ripple voltage} \times 100}{\text{output voltage}}$$

The ripple voltage shall not exceed 0.50 percent rms (1.50 percent peak-to-peak).

4.6.3.9 Output isolation. With the unit deenergized, measure the insulation resistance between each output terminal and the chassis (or cabinet) with the 500 volt dc megger (4.5.4). The resistance shall measure one megohm or greater.

4.6.3.10 Loss of phase protection. Disconnect one of the three input phases. The power supply shall stop operating. When the phase is connected again the supply shall operate as specified herein.

4.6.3.11 Current limiting circuit. With the voltage range selector in the HIGH range position, verify operation of the current limiting circuit. Increase the load current (by gradually decreasing the load resistance to zero) from 50 amperes to more than 100 amperes within a period of not more than ten seconds to determine the maximum current at which limiting action occurs. Verify that the output voltage returns to its preset value when the overload is removed.

4.6.3.11.1 With the voltage range selector in the LOW range position, verify operation of the current limiting circuit. Increase the load current (by gradually decreasing the load resistance to zero) from 50 amperes

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to mere than 200 amperes within a period of not more than ten seconds to determine the maximum current at which limiting action occurs. Verify that the output voltage returns to its preset value when the overload is removed.

4.6.3.12 Series operation. Verify that the outputs of two Type IV(A) power supplies can be connected series additive to obtain an output voltage equal to the sum of The output voltages of the individual power supplies, No interconnection except the series connection of output terminals shall be used.

NOTE

When operating two power supplies with series-connected outputs, the output voltage controls shall be varied simultaneously to approximately' balance the voltage and load between the two limits.

4.6.3.13 Verify that the power supply operates as specified for any comibination of resistive, inductive, capacitive, or rotating inverter load within load rating of the power supply. The testing of the preproduction samples for this requirement shall be accomplished using the actual weapons and test equipment specified in 3.3.2.11. After testing of the preproduction samples has confirmed that the design is compatible with the weapons and equipment, this test can be conducted by the contractor using substitute loads, the designs of which are approved by the Government procurement agency.

4.6.3.14 Verify that changes in the power supply output voltage control result in simultaneous changes in the dc output voltage, with negligible lag.

4.6.3.14.1 Verify that the voltage control is capable of adjusting the output voltage in the range of 6 to 90 volts within less than one-half volt.

4.6.3.14.2 Verify that the control of the dc output voltage is linear within plus or minus 20 degrees of shaft rotation at a minimum of 5 points distributed throughout the voltage range. Example: Divide the total mechanical degrees of shaft rotation (say 300°) by the total voltage range covered (say 5 to 95 = 90 volts) to obtain degrees per volt

$\left(\frac{300°}{90V} = 3\text{-}1/3\ °/V\right)$. An output of 50 volts shall occur at 50V x 3-1/3 °/V = 167° (167° ±20° = 147° to 187°).

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4.6.4 packaging, packing, and marking. packaging, packing, and marking shall comply with the requirements of Section 5.

4.6.5 If an equipment fails any of' the above tests, the failed equipment shall be returned to the manufacturer and shall not be resubmitted for acceptance until corrective action has been accomplished.

4.7 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.6, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Naval Ordnance Systems Command. The inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

(b) Run or test plan used.

(c) Serial or identification number of unit.

(d) Tests results of preproduction and periodic production tests.

4.7.1 Temperature. Place the power supply in a temperature chamber. With the unit operating, lower the ambient temperature to 30 degrees F. Allow the unit to stabilize at that temperature and perform the quality conformance tests of 4.6. The power supply shall meet the performance requirements of 4.6 of this specification at this temperature. Raise the ambient temperate to 130 degrees F and allow the unit to stabilize at this temperature. While operating at 130 degrees F. the power supply shall meet all the performance requirements of 4.6 of this specification Equipment should be allowed to stabilize at testing temperature for 24 hours before actual tests are begun.

4.7.2 Humidity. Operate the power supply in a humidity chamber at a relative humidity of 90 to 95 percent and a temperature of 130 degrees F for a period of 48 hours. After 48 hours of operating in this environment, the power supply shall meet the requirements cited in 4.6 of this specification. Tests are to be made while equipment is operating at 90-95 percent humidity and at a temperature of 130 degrees F.

NOTE

Only the two power supplies furnished as preproduction units are to be subjected to the rough handling tests of 4.7.3.

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4.7.3 Rough handling. The two packaged power supplies submitted as preproduction units shall be subjected to the rough handling tests of MIL-P-116. Upon completion of the rough handling tests, the power supplies shall be capable of meeting all performance requirements and product characteristics of this specification.

4.7.4 Life test. The power supply shall be subjected to 1000 hours of normal operation with only minor maintenance and repair. For the purpose of this specification, normal operation is defined as follows (see 6.2.8 regarding life test):

(a) Ambient temperature: 65 degrees to 95 degrees F.

(b) Operating 48 cycles each day as follows:

(1) 5 minutes: no load

(2) 20 minutes: 3000 watts dc load

(3) 5 minutes: off

4.7.5 Operating temperature. Operate the power supply with full load at an ambient temperature of 77 degrees F minimum until the temperature rise has stabilized at its maximum value. By means of a device such as a thermocouple, measure the temperature of the hottest exposed part, including the enclosure. This temperature shall not exceed 130 degrees F. Measure the temperature of the front panel and the operating controls; it shall not exceed 100 degrees F.

5. PREPARATION FOR DELIVERY

5.1 Unit pakaging

5.1.1 Level A. Power Supply Type IV(A) shall be packaged in accordance with Method IIa of MIL-P-116. Prior to packaging, all movable parts shall be secured and blocked to prevent movement. Material for the gaskets used to fabricate the package shall conform to MIL-G-12803, Type I, Class 2, Group 1. The adhesive used for cementing the gaskets shall conform to MIL-A-5092, Type III. The humidity indicator for the packages shall conform to Military Standard MS-20003. The package shall be provided with an inspection window conforming to MIL-W-10434, Type II.

5.1.2 Level C. Unit packaging shall be in accordance with Method III of MIL-P-116. The packaging shall be adequate to provide protection from hazards of contamination and physical damage from source of supply to the first receiving activity for immediate use.

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5.2 Packing

5.2.1 Level A. The power supply, packaged as described in 5.1.1, shall be individually packed in a nailed wood box conforming to PPP-B-621, Class 2, style 2, Type 3. The box shall be modified to include a skid base with rubbing strips and interior diagonal bracing. The box shall be provided with an inspection door tcl allow inspection of the humidity indicator inside the unit package.

5.2.2 Level C. The power supply, packaged in accordance with 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall conform to the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special marking. The shipping crate shall be marked with the word “UP” and arrows indicating the top of the crate, in accordance with MIL-STD-129.

5.3.2 Normal marking. In addition to the marking required by contract or order, the shipping crate shall be marked in accordance with the requirements of MIL-STD-329, including serial numbers furnished by the contracting agency.

6. NOTES

6.1 Intended use. Power Supply Type IV(A) is intended for use in Naval shipboard or snore-based torpedo workshops to supply dc power to torpedo electrical systems and components normally energized by storage batteries.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Title, number, and date of this specification.

6.2.2 Quantity of additional technical manuals required (25 percent).

6.2.3 Exceptions to this specification and other documents.

6.2.4 Assigned activity for preproduction and periodic production inspections.

6.2.5 Unless otherwise specified in the contract or order, power supplies shall be packaged for storage as specified in 5.1.1 and packed for storage as specified in 5.2.1. However, environmental testing of power

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supplies in the packaged and packed condition shall be accomplished only as specified by this specification and net as specified by MIL-P-116.

6.2.6 With his bid, the contractor shall submit outline drawings of the unit(s) on which he is bidding. Depicted shall be details such as size, weight, finish, general arrangement of controls and indicators, and enclosures.

6.2.7 With his bid, the contractor shall recommend and submit a list of repair parts for the power supplies. If awarded a contract to supply units, the contractor shall prepare and submit complete Naval Ordnance Systems Command drawings for all repair parts required by the procuring activity.

6.2.8 Upon award of contract and in the event that the life test of this specification (4.7.4) is waived, the contractor agrees to replace, repair, or have repaired at no cost to the Government any unit that fails to meet the operating life requirement (3.3.2.12).

6.2.9 All additional samples required because of failure of any unit(s) to pass the prescribed periodic production examinations shall be provided and inspected at the expense of the contractor.

6.2.10 Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Naval Ordnance Systems Command when testing is accomplished at other than a Government activity.

6.3 Criteria for use of proper level of preservation, packaging, and packing

6.3.1 Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.3.2 Level C. This level shall be used for those items which are to be shipped to domestic installations for immediate use at the first receiving activity.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

MIL-P-22497C(OS) Power Supply, Type IV(A) (Torpedo Testing)

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

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DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C. 20360

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